
UNIVERSITY OF BOTSWANA

2010/2011 – EXAMINATIONS

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COURSE NUMBER MAT111 DURATION 2 hrs DATE NOV 2010

TITLE OF PAPER INTRODUCTORY MATHEMATICS I

SUBJECT MATHEMATICS TITLE OF EXAMINATION BSc I

MORNING/AFTERNOON

INSTRUCTIONS:

- ANSWER ALL QUESTIONS IN SECTION A AND ANY TWO(2) QUESTIONS FROM SECTION B.
- ALL MARKS ARE INDICATED IN BRACKETS [].

NUMBER OF PAGES INCLUDING COVER PAGE: 5

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Section A

Answer All Questions: Each Question carries 10 Marks

Question 1

(a) Write $0.24\bar{1}$ in the form $\frac{p}{q}$, where $p, q \in \mathbb{Z}$ and $q \neq 0$. [3]

(b) Simplify $\frac{x^8 x^{-3}}{x^{-5} x^2}$ [3]

(c) Solve for x , $3^{x-4} = 2^{x+1}$ [4]

Question 2

Solve the following inequalities

(a) $|3x - 2| \leq 4$ [4]

(b) $2 \leq |4x + 1| < 5$ [6]

Question 3

(a) Find the domain and range of the function $f(x) = \sqrt{3x - 15}$ [5]

(b) Find the equation of the line that goes through $(-1, -2)$ and is perpendicular to $2x - y + 1 = 0$. [5]

Question 4

a) Sketch the graphs of

$$y = e^x; \quad y = 3x + 1$$

Hence solve the equation

$$e^x - 3x - 1 = 0$$

[4]

b) Find the values of α and β such that when

$$f(x) = \alpha x^3 - \beta$$

is divided by $(x - 2)$ and $(x - 1)$ the remainders are 1 and -1 respectively. [6]

Question 5

(a) Prove the following trigonometric identities:

(i) $\sin^4 \theta + \cos^4 \theta = 1 - 2 \sin^2 \theta \cos^2 \theta$

(ii) $\tan \frac{x}{2} = \frac{\sin x}{1 + \cos x}$ [3,3]

b) Show that $\sin(x + \frac{\pi}{2}) = \cos x$ using a double angle identity. [4]

Question 6

(a) Compute the complex conjugate of $-7 + 5i$ [2]

(b) Let $z = 2 + 3i$ and $w = 5 - 4i$,
write $\frac{z}{w}$ and zw in the form $a + bi$ [5,3]

Section B

Answer TWO(2) Questions OUT of FOUR:
Each Question carries 20 Marks

Question 7

- (a) Put the surd $\sqrt{8 + 2\sqrt{15}}$ in the form $\sqrt{a} + \sqrt{b}$ [6]
- (b) Solve $3^{x+1} - (2)3^{-x} + 5 = 0$ [8]
- (c) For what value(s) of k does the equation $x^2 - 3x + 2 = k(2x - 5)$ have equal roots. [6]

Question 8

- (a) Show that the function $f : \mathbb{R} \rightarrow (0, \infty)$ defined by $f(x) = e^{3x}$ is bijective. [6]
- (b) Given
$$f(x) = \frac{x^2 + 1}{2x^2 + 1},$$
find an expression for $f^{-1}(x)$ and verify that $(f^{-1} \circ f)(x) = x$. [9]
- (c) Solve the inequality $x^2 + 2x + 1 \leq 16$ [5]

Question 9

(a) (i) Sketch the graph of $f(x) = x^2 + 2x - 1$ [8]

(ii) Hence use translation on the graph in (i) above to sketch the graph of $g(x) = (x - 1)^2 + 2(x - 1)$ [4]

(b) Find the remainder when

$$f(x) = x^3 - 2x^2 - 5x + 6$$

is divided by $(x - 1)$, and hence factorize $f(x)$. [8]

Question 10

(a) Prove

$$\tan 2x = \frac{2 \tan x}{1 - \tan^2 x} \quad [5]$$

(b) Find all solutions of the following trigonometric equation

$$\cos^2 2x = \frac{1}{2} \cos 2x$$

for $0 \leq x < 2\pi$ [10]

(c) Determine the polar form of the following complex number $-1 - \sqrt{3}i$ [5]